Glaucoma Detection by Combining SLO and OCT Fundus Images: An Ensemble of Pretrained Two-Branch EfficientNets

Rushikesh

November 2026

1 Introduction

Task. Glaucoma is a chronic optic-nerve disease and a leading cause of irreversible blindness, so automated screening from retinal imaging is valuable. We address one problem from the FairDomain benchmark [1]: *given two grayscale images of the same eye, an SLO fundus image and an OCT-derived fundus image, predict whether the eye has glaucoma* (binary classification). A solution outputs a probability $P(\text{glaucoma})$ for each image pair. It is judged by the area under the ROC curve (AUC) on a fixed held-out test set and, because the data carry demographic attributes, by how evenly that performance is spread over gender, race and ethnicity groups (Section 4.3).

1.1 The data

Contents. The release holds 5,000 samples in a fixed split of 3,000 training and 2,000 test samples. Each sample is one `.npz` file with the two images (`slo_fundus` and `oct_fundus`; each 224×224, 8-bit, single channel), the label `glaucoma` (0 or 1), and patient attributes: age, gender, race, ethnicity, language, marital status and the visual-field mean deviation (MD). A CSV file indexes the split and the attributes. No segmentation masks are included, so the task is classification only.

What the images look like. Figure 1 shows random samples of each class. Both modalities show the optic disc and the retinal vessels around it. The SLO fundus image is darker, with uneven illumination and dark patches; the OCT fundus image is brighter and smoother. The two modalities therefore differ clearly in appearance, but *the two classes do not look different to the eye*, and simple intensity statistics do not separate them either: over 300 random training images the mean pixel value is 73 (no glaucoma) against 70 (glaucoma) for SLO, and 169 against 169 for OCT. The signal is subtle structural information that a model has to learn.

Labels and demographics. Glaucoma is the positive class in 44% of both splits (1,317 of 3,000 training and 887 of 2,000 test samples), a mild imbalance that is one reason we report AUC rather than accuracy. The demographic groups are strongly imbalanced (Table 1): 73% of samples are female, 77% white, 16% Black, 8% Asian and 4% Hispanic. Only 83 test images belong to Hispanic patients, so any estimate for that group is very noisy.

Clinical variables. MD summarises a visual-field test (in dB; 0 is normal, more negative means more field loss). It averages 0.1 ± 0.8 in non-glaucoma and -8.2 ± 5.7 in glaucoma training samples, so it almost encodes the label: giving it to the model raises test AUC to 0.958 (Appendix F). We treat it as a leakage risk and use it only in one clearly labelled upper-bound run, never in a reported

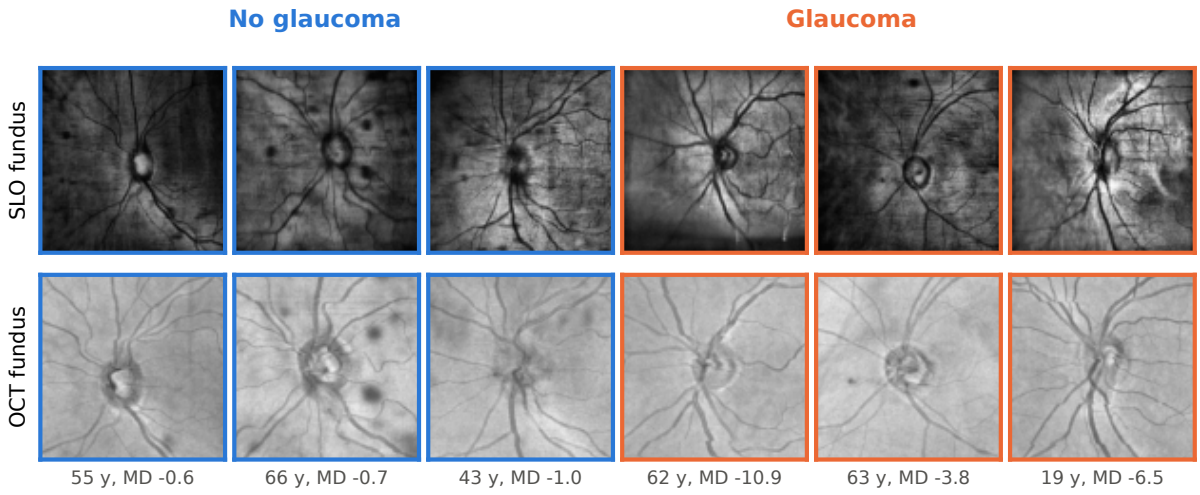


Figure 1: Randomly chosen training samples (seed 0, not hand-picked) for each class. Top row: SLO fundus; bottom row: OCT fundus. Under each pair: the patient's age and visual-field mean deviation (MD, dB). Blue frame: no glaucoma; orange frame: glaucoma.

Split	N	Glaucoma	Female	White	Black	Asian	Hispanic
Training	3000	1317 (44%)	2204 (73%)	2308 (77%)	467 (16%)	225 (8%)	121 (4%)
Testing	2000	887 (44%)	1449 (72%)	1522 (76%)	301 (15%)	177 (9%)	83 (4%)

Table 1: Dataset composition. Percentages are of each split.

comparison. Age differs by class (57.8 against 64.4 years on average) but adding it did not help (Appendix F).

What an engineer should keep in mind.

- **Small data:** after holding out validation data only 2,550 training images remain, so overfitting is the main difficulty (Appendix C).
- **Small test set:** 2,000 images give AUC confidence intervals of about ±0.02; differences smaller than that between two models are not meaningful.
- **No official validation set:** we hold out a stratified 15% of the training images for all model and configuration selection (Section 4.1).
- **Two different views:** a model trained on one modality transfers poorly to the other (Appendix F), so the modalities must be handled separately and combined.
- **Fairness-aware evaluation:** results should be reported per demographic group, not only overall.

Our approach and result. Our final model is an *ensemble of eight ImageNet-pretrained two-branch EfficientNet-B0 networks*, one branch per modality, trained with five-fold cross-validation plus three runs on all training data, with horizontal-flip test-time averaging. It reaches a test AUC of 0.834 (95% CI 0.816–0.852) against 0.717 for a from-scratch ResNet18 baseline. The main

text describes only this model and the experiments that justify it; the other experiments (a 13-model backbone comparison, a training-recipe study with ablations, fusion variants, a foundation model, fairness losses, cross-domain transfer) are documented with their motivation and setup in the appendices.

Contributions. (1) A simple, strong pipeline for SLO+OCT glaucoma classification with bootstrap confidence intervals and per-group results. (2) A controlled demonstration, on the same backbone and recipe, that using both modalities beats either alone (0.827 against 0.758 and 0.764). (3) Evidence that averaging models trained on different data splits gains about 0.03 AUC over its members. (4) An honest account, in the appendices, of what did *not* help, including a hypothesis of ours that the ablations refuted.

2 Related Work

- **Datasets from the Harvard Ophthalmology AI Lab.**
  - Harvard Glaucoma Fairness [2]: glaucoma data with 2D and 3D images. It proposes Fair Identity Normalization (FIN), which makes feature importance equal across groups.
  - Harvard Eye Fairness [3]: 30,000 subjects with 3D OCT. It proposes Fair Identity Scaling (FIS), which weights the loss by individual and by group.
  - FairCLIP [4]: fairness for vision–language models on related data.
  - FairDomain [1] (our data): fairness under domain shift, for segmentation and classification. It treats image types as domains and proposes a fair identity attention (FIA) module.
- **Fairness metrics.** This line of work commonly reports group-wise AUC, worst-group AUC and equity-scaled AUC (ES-AUC).
- **Recent glaucoma work (not reproduced by us).**
  - FairEnc [5]: fair vision–language model on SLO images with clinical notes.
  - Fair-Eye Net [6]: combines fundus photos, OCT-derived metrics and visual-field indices, with a fairness constraint.
  - RED-Sphere [7]: domain generalisation across populations for fundus disease.
- **Models and tools we use.**
  - Final model: EfficientNet [9] with ImageNet [19] weights from `timm` [20].
  - Also tested (Appendix B): ResNet [8], DenseNet [10], RegNet [11], ConvNeXt [12], ViT [13], DeiT [14], Swin [15], CoAtNet [16], MaxViT [17], DINOv2 [18].
  - Training: AdamW [22], one-cycle schedule [23], label smoothing [27], drop-path [26]. Appendices only: mixup [24], CutMix [25], gradient reversal [28].

3 Method

3.1 Problem

- Input: a pair $(\mathbf{x}_S, \mathbf{x}_O)$ of 224×224 single-channel images (SLO fundus, OCT fundus).

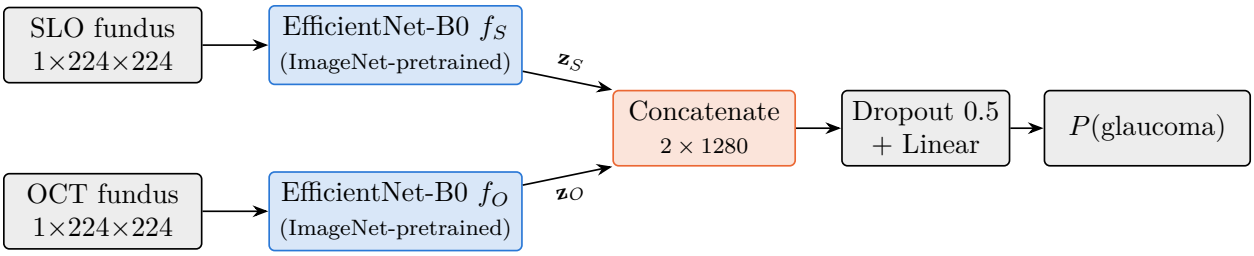


Figure 2: One ensemble member: a two-branch late-fusion network with an EfficientNet-B0 backbone per modality.

- Output: $p(y = 1 \mid \mathbf{x}_S, \mathbf{x}_O)$, the probability of glaucoma ($y \in \{0, 1\}$).
- Demographic attributes g are used only to measure fairness.

3.2 Two-branch EfficientNet-B0

- One EfficientNet-B0 backbone per modality (f_S, f_O). Separate weights, initialised from ImageNet-1k, 4.0 M parameters each (Figure 2).
- The pooled features (1280 values per branch) are concatenated, then dropout (0.5), then one linear layer with two outputs.
- `timm` adapts the pretrained first layer to one input channel.
- Pixels are scaled to $[0, 1]$, then normalised as $(x - 0.5)/0.25$.
- Other fusion designs (early, gated, cross-attention) did not clearly help (Appendix D).

3.3 Training recipe

- Loss: cross-entropy with label smoothing 0.1.
- Optimiser: AdamW with a one-cycle schedule. Peak learning rate 5×10^{-4} (5× larger for the new head), weight decay 10^{-2}, batch size 32, 25 epochs, gradient clipping 1.0.
- Regularisation: drop-path 0.2, head dropout 0.5, and an exponential moving average (EMA) of the weights (decay 0.995) used for evaluation.
- Augmentation (on GPU): horizontal flip, rotation ±15°, scale ±10%, shift up to 5% of the image size, brightness and contrast jitter.
- The best epoch (by validation AUC) is kept.
- This recipe was picked automatically, by validation AUC, from twelve runs of the recipe study (Appendix C). Its validation AUC was 0.852, against 0.832 for the plain recipe (mean of 3 seeds). The gain was larger than the seed-to-seed spread.

3.4 Ensemble and test-time augmentation

- The final prediction is the average of the probabilities of 8 models (Figure 3):
  - **5 cross-validation members.** The 3,000 training images are split into 5 stratified folds. Each member trains on 4 folds (2,400 images) and uses the held-out fold (600 images) to choose its best epoch.

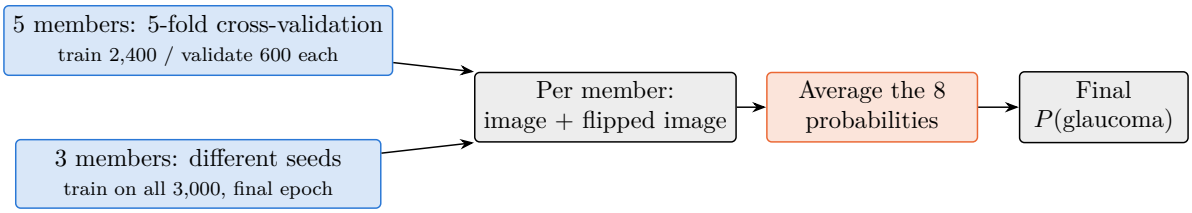


Figure 3: The final ensemble: eight EfficientNet-B0 two-branch models, each evaluated on an image and its mirror image, with probabilities averaged.

  - **3 full-data members.** Three seeds, trained on all 3,000 images, with no validation set. The final-epoch weights are used.

- Test-time augmentation (TTA): each member's prediction is averaged with its prediction on the horizontally flipped image.
- The ensemble was defined before any test result was seen. It is one of four ensembles evaluated in sweep 3 (Appendix G).

4 Experiments

4.1 Data splits

- The data are described in Section 1.1 (Table 1).
- Validation set: a stratified 15% of the 3,000 training images (450). It is used to choose epochs and configurations.
- Single-run models train on the other 2,550 images.
- Cross-validation members use 5 stratified folds of the 3,000 images. Full-data members use all 3,000.
- The 2,000 test images are never used for training or for automatic selection.

4.2 Implementation and hyper-parameters

- PyTorch [21] with `timm` [20]; fp32; one Tesla P100 (16 GB). One member trains in about 11–12 minutes.
- Hyper-parameters are those of Section 3.3 (Table 2).

4.3 Evaluation protocol

- Main metric: AUC on the 2,000 test images. It needs no threshold and is not affected by the 44/56 class balance.
- Uncertainty: bootstrap 95% confidence interval over test images (1,000 resamples). Intervals are about ± 0.02 wide. Differences smaller than that are not meaningful.
- Fairness metrics:
  - AUC inside each gender, race and ethnicity group.

Hyper-parameter	Value
Backbone	EfficientNet-B0 ×2 (ImageNet-1k, `ra_in1k` weights)
Optimiser / schedule	AdamW, one-cycle (15% warm-up), gradient clip 1.0
Learning rate (body / head)	5×10^{-4} / 2.5×10^{-3}
Weight decay, batch size, epochs	10^{-2}, 32, 25
Label smoothing, drop-path, head dropout	0.1, 0.2, 0.5
EMA decay	0.995
Augmentation	flip, rotation ±15°, scale ±10%, shift ≤ 5%, brightness/contrast jitter
Test-time augmentation	horizontal flip

Table 2: Hyper-parameters of every ensemble member.

Model	Input	Test AUC	95% CI	+ flip TTA	ES-AUC race	Worst-group AUC
ResNet18, two-branch, from scratch (baseline)	SLO+OCT	0.717	[0.694, 0.739]	0.725	0.696	0.706
EfficientNet-B0, SLO only	SLO	0.758	[0.736, 0.778]	0.762	0.727	0.723
EfficientNet-B0, OCT only	OCT	0.764	[0.742, 0.784]	0.771	0.750	0.753
EfficientNet-B0, two-branch, plain recipe (one run)	SLO+OCT	0.827	[0.808, 0.844]	0.828	0.815	0.816
EfficientNet-B0, two-branch, final recipe (one run)	SLO+OCT	0.817	[0.797, 0.835]	0.820	0.793	0.803
five cross-validation members, mean ± s.d.	SLO+OCT	0.800 ± 0.008	–	–	–	–
Ensemble of the 5 cross-validation models	SLO+OCT	0.830	[0.811, 0.848]	0.831	0.807	0.820
Final: ensemble of 8 models	SLO+OCT	0.834	[0.816, 0.852]	0.835	0.802	0.821

Table 3: Main results on the 2,000 test images. "Plain recipe" = flips and mild jitter, 15 epochs. "Final recipe" = Section 3.3. TTA = horizontal-flip test-time averaging.

  - Worst-group AUC: the lowest AUC over gender and race groups with at least 100 test samples.
  - Equity-scaled AUC (ES-AUC), computed per attribute:

$$\text{ES-AUC} = \frac{\text{AUC}}{1 + \sum_g |\text{AUC} - \text{AUC}_g|}. \tag{1}$$

- ES-AUC is our own implementation of the equity-scaled AUC used in the Harvard fairness benchmarks. Absolute values may differ from the original definition.

4.4 Baselines

- **ResNet18 from scratch:** same two-branch design, random initialisation, plain recipe, 15 epochs.
- **Single-modality EfficientNet-B0:** same backbone and plain recipe, SLO only or OCT only.
- **Single two-branch EfficientNet-B0:** one run with the plain recipe (from our first backbone comparison) and one run with the final recipe.
- **Cross-validation members:** the five individual members. They measure the gain from ensembling on identical models.

4.5 Results

Table 3 and Figure 4 show the main results.

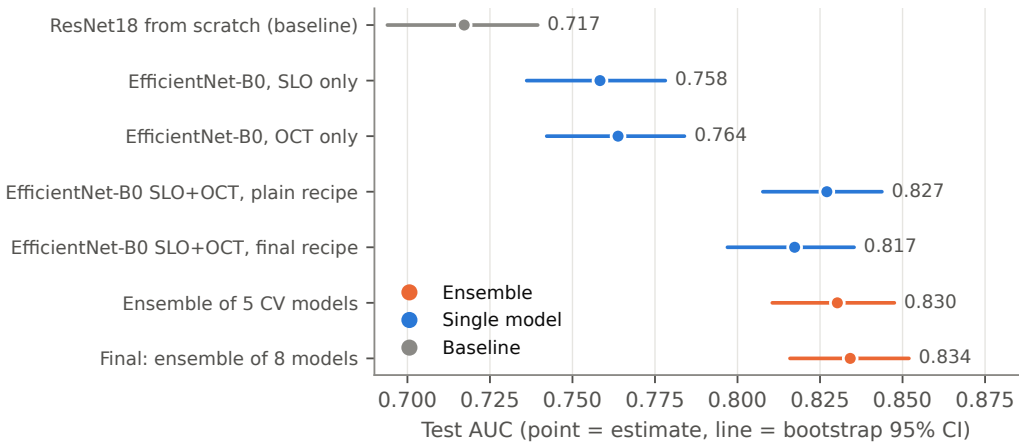


Figure 4: Test AUC with bootstrap 95% confidence intervals for the baseline, single models and ensembles.

- **Both modalities matter.**
  - Same backbone and recipe: SLO only 0.758, OCT only 0.764, both 0.827.
  - The gain (+0.06 to +0.07) is well outside the confidence intervals.
  - The two images carry different information: a model trained on one modality transfers poorly to the other (Appendix F).
- **Pretrained EfficientNet versus the baseline.**
  - From-scratch ResNet18: 0.717. Pretrained two-branch EfficientNet-B0, single runs: 0.817–0.827.
  - This gap mixes pretraining and architecture. Every pretrained backbone we tried beat the scratch baseline (Appendix B), but we did not train scratch versions of the other backbones. So we cannot credit pretraining alone.
- **Ensembling.**
  - The five cross-validation members score 0.800 ± 0.008 each (range 0.789–0.808). Their average scores 0.830: a gain of 0.030 on otherwise identical models.
  - Adding the three full-data members gives the final 0.834 [0.816, 0.852]. Flip TTA adds 0.001.
  - The ensemble is not clearly better than the best single run (0.827, plain recipe, one seed). The difference is well inside the confidence intervals.
  - The benefit is stability: the ensemble does not depend on one lucky run. The 0.827 run may be partly luck (Appendix C).
- **Fairness across groups** (Table 4).
  - Final ensemble: AUC 0.82–0.86 across gender and race groups. Worst-group AUC 0.821 (baseline: 0.706). ES-AUC for race: 0.802.
  - The Hispanic group (83 test images) has the lowest AUC (0.806) but a very wide confidence interval. We make no claim about it.

n (test)	Female 1449	Male 551	Asian 177	Black 301	White 1522	Non-Hisp. 1917	Hispanic 83
ResNet18 from scratch (baseline)	0.721	0.706	0.711	0.735	0.710	0.723	0.556
EfficientNet-B0, final recipe (one run)	0.816	0.822	0.831	0.803	0.814	0.819	0.811
Final: ensemble of 8 models	0.832	0.841	0.856	0.821	0.829	0.836	0.806

Table 4: Test AUC within demographic groups.

  - Fairness-specific losses did not help (Appendix E).

4.6 Limitations

- The test set has 2,000 images. Confidence intervals are about ±0.02, and the ensemble's lead over the best single model is inside them.
- We evaluated four ensemble compositions and report the one with the highest test AUC as the final model. Its figure is mildly optimistic (the four ranged from 0.816 to 0.834; Appendix G).
- Automatic selection used validation data only, but we read test results while planning later experiments.
- Appendix C found that flips alone (no rotation, scale or shift) raise single-model AUC by about 0.02. The final ensemble does not use this, because the ablation ran after the ensemble was trained.
- ES-AUC is our own implementation.
- We did not implement the published FIN, FIS and FIA methods and did not compare with published numbers.

5 Conclusions

- **Result.** An ensemble of eight ImageNet-pretrained two-branch EfficientNet-B0 networks reaches 0.834 test AUC (95% CI 0.816–0.852). The from-scratch ResNet18 baseline reaches 0.717.
- **What mattered.**
  - Using both modalities: about +0.06 to +0.07 AUC over either alone.
  - Averaging models trained on different splits: about +0.03 over its members.
  - Performance is similar across gender and race groups.
- **Strengths.** Simple: standard pretrained backbones, concatenation fusion, probability averaging.
- **Weaknesses.**
  - The small test set cannot separate the ensemble from the best single model.
  - Eight networks cost eight times as much to run.
  - The best recipe we found (flip-only augmentation) came too late to be included.

- **Future work.**
  - Combine the helpful recipe changes and test them with several seeds.
  - Implement FIN, FIS and FIA faithfully and compare on the full dataset.
  - Fine-tune retinal foundation models.
  - Use the two image types as source and target domains with domain adaptation, since transfer between them is poor.
  - Use cross-validation over the full data to reduce noise from a single test split.

References

[1] Y. Tian, C. Wen, M. Shi, M. M. Afzal, H. Huang, M. O. Khan, Y. Luo, Y. Fang, and M. Wang. FairDomain: Achieving fairness in cross-domain medical image segmentation and classification. In *ECCV*, 2024. arXiv:2407.08813.

[2] Y. Luo et al. Harvard Glaucoma Fairness: A retinal nerve disease dataset for fairness learning and fair identity normalization. *IEEE Trans. Medical Imaging*. arXiv:2306.09264.

[3] Y. Luo, Y. Tian, M. Shi, T. Elze, and M. Wang. Harvard Eye Fairness: A large-scale 3D imaging dataset for equitable eye diseases screening and fair identity scaling. arXiv:2310.02492, 2023.

[4] Y. Luo et al. FairCLIP: Harnessing fairness in vision-language learning. In *CVPR*, 2024. arXiv:2403.19949.

[5] FairEnc: A fair vision-language model with fair vision and text encoders for glaucoma detection. arXiv:2605.04882.

[6] Fair-Eye Net: A fair, trustworthy, multimodal integrated glaucoma full chain AI system. arXiv:2601.18464.

[7] RED-Sphere: Hyperspherical residual edge debiasing for cross-population fundus disease domain generalization. arXiv:2607.10777.

[8] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, 2016.

[9] M. Tan and Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *ICML*, 2019.

[10] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger. Densely connected convolutional networks. In *CVPR*, 2017.

[11] I. Radosavovic, R. P. Kosaraju, R. Girshick, K. He, and P. Dollár. Designing network design spaces. In *CVPR*, 2020.

[12] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie. A ConvNet for the 2020s. In *CVPR*, 2022.

[13] A. Dosovitskiy et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

[14] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou. Training data-efficient image transformers and distillation through attention. In *ICML*, 2021.

[15] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin Transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, 2021.

[16] Z. Dai, H. Liu, Q. V. Le, and M. Tan. CoAtNet: Marrying convolution and attention for all data sizes. In *NeurIPS*, 2021.

[17] Z. Tu, H. Talebi, H. Zhang, F. Yang, P. Milanfar, A. Bovik, and Y. Li. MaxViT: Multi-axis vision transformer. In *ECCV*, 2022.

[18] M. Oquab et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

[19] O. Russakovsky et al. ImageNet large scale visual recognition challenge. *IJCV*, 2015.

[20] R. Wightman. PyTorch Image Models (timm). `https://github.com/huggingface/pytorch-image-models`, 2019.

[21] A. Paszke et al. PyTorch: An imperative style, high-performance deep learning library. In *NeurIPS*, 2019.

[22] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *ICLR*, 2019.

[23] L. N. Smith and N. Topin. Super-convergence: Very fast training of neural networks using large learning rates. In *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications*, SPIE, 2019.

[24] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz. mixup: Beyond empirical risk minimization. In *ICLR*, 2018.

[25] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo. CutMix: Regularization strategy to train strong classifiers with localizable features. In *ICCV*, 2019.

[26] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Q. Weinberger. Deep networks with stochastic depth. In *ECCV*, 2016.

[27] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the Inception architecture for computer vision. In *CVPR*, 2016.

[28] Y. Ganin and V. Lempitsky. Unsupervised domain adaptation by backpropagation. In *ICML*, 2015.

A Overview of all experiments

- Besides the final model we ran four batches of experiments: 51 training runs plus ensembles.
- All runs use the same data, the same split and the same validation protocol (Section 4.1). The epoch is chosen on validation AUC. Nothing was selected automatically on test AUC.
- Table 5 lists the batches. The next appendices give the details.

B Backbone and modality comparison (sweep 1)

- **Motivation.** Find out how much pretraining, the architecture family (CNN, transformer, hybrid) and the use of both modalities matter.
- **Setup.**
  - 13 configurations, *plain* recipe: cross-entropy, AdamW, one-cycle, 15 epochs, flips and mild brightness jitter. Learning rates: Table 6.

Appendix	Question (motivation)	Setup	Main outcome
B Backbones	Which architecture family and modality choice work best?	13 configurations, plain recipe, 15 epochs	Pretrained 0.747–0.827 vs scratch 0.717; EfficientNet-B0 best (0.827); both modalities help
C Training recipe	How to fight overfitting on 2,550 images?	Stronger recipe on Swin-Tiny and EfficientNet-B0; mixup, drop-path, LR, schedule; 4 ablations	Drop-path helps (0.817 vs 0.798); strong geometric augmentation hurts (flip-only 0.819)
D Fusion / foundation	Is smarter fusion or a self-supervised ViT better?	Gated fusion, cross-attention, DINOv2-S, Swin-scale recipe	No gain: 0.812, 0.795, 0.809 vs 0.808 ± 0.004
E Fairness losses	Can fairness-aware training raise worst-group performance?	Group reweighting, FIS-like loss, adversarial race head	No consistent gain (0.787, 0.806, 0.805)
F Domain shift, tabular	How different are the two modalities? Do clinical variables help?	Train on one modality, test on the other; add age / MD	Transfer 0.68–0.70; age no gain; MD 0.958 (not comparable)
G Ensembles	Does averaging help, and which members?	Eight ensemble compositions, with and without TTA	0.816–0.834

Table 5: The experiments behind and around the final model.

  - Late fusion (one backbone per modality), unless marked "early" (2-channel input to one backbone). Single-modality runs use one branch.
  - Epoch chosen by validation AUC.

- **Results** (Table 11, Figure 5).
  - Every pretrained model beat the from-scratch ResNet18 (0.717): EfficientNet-B0 0.827, Swin-Tiny 0.807, DenseNet121 0.803, CoAtNet-0 0.793, ViT-Small 0.790, MaxViT-Tiny 0.778, DeiT-Small 0.768, ResNet50 0.768 (early fusion 0.747), ConvNeXt-Tiny 0.767.
  - Models between 0.79 and 0.83 have almost fully overlapping confidence intervals. No architecture family is clearly best.
  - Modality (EfficientNet-B0, Figure 6): SLO only 0.758, OCT only 0.764, both 0.827.
  - ResNet50: late fusion 0.768, early fusion 0.747 (inside the intervals).
  - Swin-Tiny had the best *validation* AUC (0.844) but not the best test AUC. The 450-image validation set is noisy, so the recipe study (Appendix C) uses several seeds.

- **Takeaway.** Pretrained two-modality models work. The architecture family is secondary. We chose EfficientNet-B0: best result and cheapest strong model (about 5 minutes per run).

C Training recipe study and ablations (sweeps 2–3)

- **Motivation.** All models overfit. The plain EfficientNet-B0 reaches a training loss of about 0.008 while validation AUC stops improving early (Figure 7). We tried to regularise harder.

- **Setup.**
  - *Stronger recipe*: on-GPU rotation $\pm 15°$, scale $\pm 10\%$, shift $\leq 5\%$; label smoothing 0.1; EMA of the weights; $5\times$ learning rate on the new head; 25 epochs.

Backbone	Peak LR	Batch size	Weight decay
ResNet18 (scratch)	3e-4	64	1e-2
ResNet50	2e-4	32	1e-2
EfficientNet-B0	5e-4	32	1e-2
DenseNet121	2e-4	32	1e-2
ConvNeXt-Tiny	1e-4	32	5e-2
ViT-S / DeiT-S / Swin-T	5e-5	32	5e-2
CoAtNet-0	1e-4	32	5e-2
MaxViT-Tiny	5e-5	16	5e-2

Table 6: Hyper-parameters of the first sweep.

  - First on Swin-Tiny (best validation AUC in sweep 1), 3 seeds.
  - Then on EfficientNet-B0 with variants: 12 epochs; mixup/CutMix; drop-path 0.2 with head dropout 0.5; lower learning rate (2×10^{-4}); gated fusion. Plus four other backbones: EfficientNet-B2, B3, RegNetY-016, DenseNet121.
  - Rule for the final configuration (validation AUC only): a variant replaces the plain stronger recipe only if its validation AUC beats the 3-seed mean by more than max(0.005, seed s.d.). Drop-path won: 0.852 against a mean of 0.832, margin 0.013.

- **Results** (Table 12 has the full list).
  - Swin-Tiny: unchanged. 0.808 ± 0.004 over 3 seeds, against 0.807 in sweep 1.
  - EfficientNet-B0: *dropped* from 0.827 to 0.798 ± 0.003.
  - Drop-path with dropout helped: 0.817.
  - No help: mixup/CutMix 0.800, lower learning rate 0.790, 12-epoch schedule 0.777.
  - Other backbones were no better than EfficientNet-B0: B2 0.796, B3 0.779, RegNetY-016 0.807, DenseNet121 0.781.

- **Ablation: why did the recipe hurt EfficientNet?** (Table 7; one seed each)
  - Our first guess: the EMA code copies batch-norm statistics from the live model instead of averaging them. That could hurt batch-norm CNNs but not the layer-norm Swin.
  - EMA off: AUC fell to 0.786. The guess was *refuted*; EMA helps.
  - Flips and mild jitter only (no rotation, scale, shift): 0.819, up from 0.798. This is about 7 seed standard deviations. Strong geometric augmentation is the main culprit.
  - Head learning rate 1× instead of 5×: 0.807, a smaller gain.
  - All three choices reverted (approximate sweep-1 setup): 0.811, below the original 0.827. So part of that 0.827 is probably single-seed luck.
  - Possible reason for the augmentation result (not tested): the images may be spatially registered, so rotations and shifts could remove position-dependent cues.
  - Not run: the combination of the helpful changes (flip-only augmentation, drop-path, head learning rate 1×). It is the most promising next experiment.

- **Takeaway.** Most standard regularisers did nothing. Drop-path was the one clear win. Geometric augmentation was harmful.

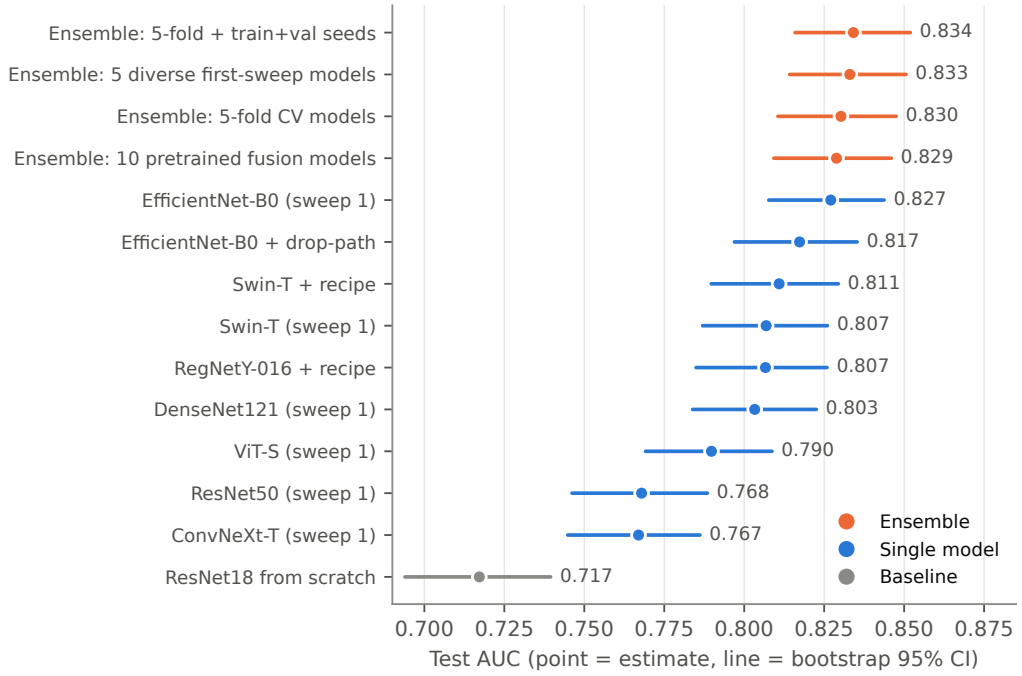


Figure 5: Test AUC with bootstrap 95% confidence intervals for the main single models and the best ensembles.

EfficientNet-B0 variant	Val AUC	Test AUC	TTA AUC
Recipe (3 seeds, mean ± s.d.)	0.832	0.798 ± 0.003	0.803
EMA off	0.817	0.786	0.792
Head LR ×1 (instead of ×5)	0.828	0.807	0.807
Flip + mild jitter only (no rotate/scale/shift)	0.853	0.819	0.824
All three reverted, no label smoothing, 15 epochs (approx. sweep-1 setup)	0.844	0.811	0.814
Sweep-1 run (original code, one seed)	0.833	0.827	0.828

Table 7: Single-factor ablations of the stronger recipe (EfficientNet-B0, one seed each; the recipe row is the three-seed mean). The last row is the original first-sweep run.

D Fusion variants and a foundation model (sweep 2)

- **Motivation.** Plain concatenation may not be the best way to combine two modalities. A large self-supervised model may transfer better than ImageNet weights.
- **Setup** (Swin-Tiny recipe, 25 epochs; Figure 8).
  - *Gated fusion* with modality dropout: $\mathbf{z} = \sigma(W[\mathbf{p}_S; \mathbf{p}_O]) \odot \mathbf{p}_S + (1 - \sigma(\cdot)) \odot \mathbf{p}_O$, with $\mathbf{p} = W'\mathbf{z}$. One modality's feature is set to zero for 20% of training samples.
  - *Cross-attention fusion* with ViT-Small: each modality's tokens attend to the other's tokens, then the class tokens are concatenated.
  - *DINOv2-Small* [18], fine-tuned, with late fusion.
- **Results** (Table 9).
  - Gated fusion 0.812, cross-attention 0.795, DINOv2 0.809. Plain concatenation: 0.808 ±

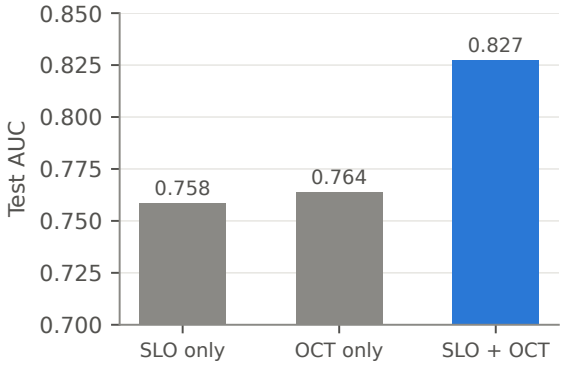


Figure 6: Contribution of each modality (EfficientNet-B0, plain recipe).

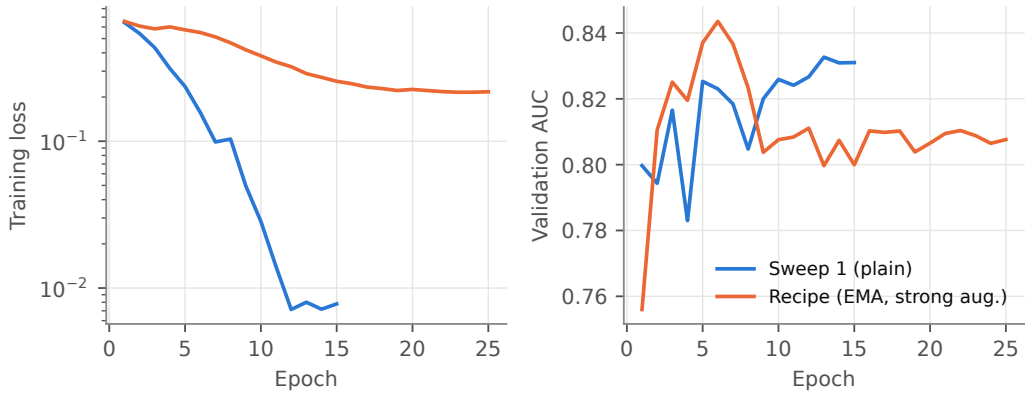


Figure 7: EfficientNet-B0: training loss (log scale) and validation AUC. The stronger recipe's loss stays higher because of label smoothing and augmentation, but its validation AUC peaks around epoch 6 and then declines.

0.004.

- No variant is distinguishable from the baseline.
- Gated fusion on EfficientNet-B0: 0.810, against 0.798 for concatenation.

- **Takeaway.** Plain concatenation is enough at this data size. The final model keeps it.

E Fairness-aware training and per-group results (sweep 2)

- **Motivation.** The data are demographically imbalanced, and fairness-aware losses are a main theme of the dataset papers [2, 3, 1].
- **Setup** (Swin-Tiny recipe).
  - *Group reweighting*: each sample is weighted by the inverse frequency of its (race, label) cell.
  - *FIS-like loss*: weight = (sample loss / batch mean loss) × (its race group's mean loss / batch mean loss), clipped to $[0.2, 5]$ and renormalised. This is our approximation of the

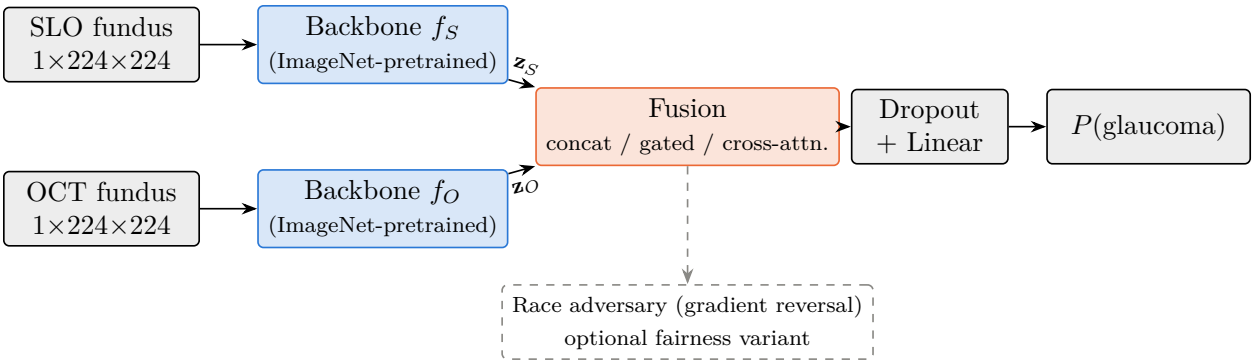


Figure 8: Variants evaluated in the appendices: the fusion block is replaced by gated or cross-attention fusion; a single backbone with a 2-channel input (early fusion) or with one modality only; an adversarial race head on the fused feature.

idea of Fair Identity Scaling [3], not an exact copy.

- *Adversarial debiasing*: a small classifier predicts race from the fused feature through a gradient-reversal layer [28] (loss weight 0.5).
- The adversarial variant was repeated on EfficientNet-B0.

- **Results** (Table 9, Table 8).
  - FIS-like 0.806 and adversarial 0.805: within noise of the plain recipe (0.808). Group reweighting: 0.787, lower.
  - EfficientNet-B0: adversarial 0.802, against 0.817 for the configuration it extends.
  - Gender and race groups differ by about 0.01–0.03 AUC for most models.
  - The Hispanic group (83 test images) varies most (0.74–0.81). Sampling noise could explain this.
- **Takeaway.** No fairness-specific loss gave a reliable benefit. The best group-level behaviour came from ensembling.

n (test)	Female 1449	Male 551	Asian 177	Black 301	White 1522	Non-Hisp. 1917	Hispanic 83
EfficientNet-B0 (sweep 1)	0.824	0.835	0.823	0.816	0.827	0.830	0.740
Swin-T + recipe (sweep 2)	0.813	0.806	0.832	0.811	0.806	0.812	0.782
EfficientNet-B0 + drop-path (sweep 3)	0.816	0.822	0.831	0.803	0.814	0.819	0.811
5-model ensemble (sweep 1)	0.833	0.835	0.829	0.826	0.830	0.834	0.796
8-model ensemble (sweep 3)	0.832	0.841	0.856	0.821	0.829	0.836	0.806

Table 8: Test AUC within demographic groups for representative models and ensembles.

F Domain shift and tabular inputs (sweep 2)

- **Motivation.**

  - FairDomain treats image types as domains [1]. If SLO and OCT carry different information, a model trained on one should transfer poorly to the other. That would explain why fusion helps.
  - Do clinical variables add anything?

- **Setup.**
  - *Cross-domain*: one Swin-Tiny branch is trained on one modality only. It is tested on the same modality and on the other (fed through the same single-channel input).
  - *Tabular*: age (and, as an upper bound, the visual-field mean deviation MD) goes through a small MLP and is concatenated to the fused feature.

- **Results** (Table 9).
  - Trained on SLO: 0.794 on SLO, 0.677 on OCT.
  - Trained on OCT: 0.786 on OCT, 0.701 on SLO.
  - Age: 0.808, no change.
  - MD: 0.958. MD is a clinical measure closely tied to the label. This is not a fair comparison and is excluded from all other comparisons.

- **Takeaway.** The two modalities are complementary, and transfer between them is an open problem. Age adds nothing.

Variant (Swin-Tiny backbone)	Val AUC	Test AUC	95% CI	ES-AUC race	Worst-group	Other-domain AUC
+ age and MD (upper bound)	0.963	0.958	[0.950, 0.965]	0.938	0.941	–
Gated fusion + modality dropout	0.842	0.812	[0.793, 0.832]	0.800	0.803	–
Swin-T + recipe	0.834	0.811	[0.790, 0.831]	0.790	0.806	–
Swin-T + recipe (seed 2)	0.836	0.810	[0.789, 0.830]	0.783	0.793	–
DINOv2-S fine-tune	0.837	0.809	[0.788, 0.829]	0.760	0.786	–
+ age as input	0.840	0.808	[0.786, 0.828]	0.785	0.802	–
FIS-like loss	0.847	0.806	[0.784, 0.826]	0.779	0.800	–
Adversarial race debiasing	0.850	0.805	[0.782, 0.823]	0.788	0.801	–
Swin-T + recipe (seed 1)	0.843	0.803	[0.782, 0.823]	0.783	0.797	–
ViT-S cross-attention fusion	0.844	0.795	[0.773, 0.815]	0.772	0.786	–
Train SLO → test OCT	0.796	0.794	[0.773, 0.812]	0.759	0.786	0.677
Group-reweighted loss	0.819	0.787	[0.764, 0.808]	0.760	0.780	–
Train OCT → test SLO	0.822	0.786	[0.765, 0.804]	0.758	0.779	0.701

Table 9: Sweep 2: stronger recipe on Swin-Tiny, 25 epochs. The "+ age and MD" row is an upper bound and not a fair comparison.

G Ensembles and test-time augmentation

- **Motivation.** Single runs overfit and are noisy. Averaging models reduces variance.
- **Setup.** Probability averaging of:
  - Sweep-1 checkpoints: five diverse models, ten pretrained fusion models, the top three by validation AUC, a greedy selection on validation AUC.
  - EfficientNet-B0 models of the final configuration: the five cross-validation members; three seeds on all 3,000 images; both together (the final model); the top three configurations of the recipe study.

  - Members were chosen in advance or by validation AUC. Each ensemble is also reported with flip TTA.

- **Results** (Table 10).
  - Test AUC ranges from 0.816 to 0.834.
  - The five diverse sweep-1 models (0.833) and the final eight-model ensemble (0.834) are tied inside the intervals.
  - Flip TTA adds 0.001 to 0.010.
  - The three full-data members score 0.791–0.794 alone, no better than single models trained on 2,550 images. More data did not help single models, but it helped the ensemble slightly (0.830 to 0.834).

- **Takeaway.** Ensembling is the most reliable gain (about +0.03 over the members). The exact members matter little. We chose the eight-member ensemble because it uses one validation-selected configuration. It had the highest test AUC of the compositions tried, so its figure is mildly optimistic.

Ensemble (probability average)	Source	n	Test AUC	95% CI	TTA AUC	ES-AUC race
5-fold + train+val seeds (8 models)	Sweep 3	8	0.834	[0.816, 0.852]	0.835	0.802
5 diverse first-sweep models	Sweep 1 checkpoints	5	0.833	[0.814, 0.851]	0.834	0.821
5-fold CV models (drop-path config)	Sweep 3	5	0.830	[0.811, 0.848]	0.831	0.807
greedy selection on validation (8 slots)	Sweep 1 checkpoints	8	0.830	[0.811, 0.847]	0.831	0.814
10 pretrained fusion models	Sweep 1 checkpoints	10	0.829	[0.809, 0.846]	0.829	0.821
top-3 of sweep-3 tiers 1–2 by validation	Sweep 3	3	0.823	[0.803, 0.841]	0.824	0.799
top-3 by validation AUC	Sweep 1 checkpoints	3	0.819	[0.800, 0.837]	0.822	0.805
3 seeds trained on train+val	Sweep 3	3	0.816	[0.797, 0.834]	0.821	0.764

Table 10: Probability-averaging ensembles. Members were chosen in advance or by validation AUC.

H Full result tables

Model	Type	Params (M)	Val AUC	Test AUC	TTA AUC	Acc.	Worst-group AUC
EfficientNet-B0	CNN	8.0	0.833	0.827	0.828	0.763	0.816
Swin-Tiny	Transformer	55.0	0.844	0.807	0.811	0.739	0.796
DenseNet121	CNN	13.9	0.837	0.803	0.810	0.740	0.794
CoAtNet-0	Hybrid	53.3	0.835	0.793	0.796	0.716	0.787
ViT-Small	Transformer	42.9	0.841	0.790	0.792	0.724	0.781
MaxViT-Tiny (early fusion)	Hybrid	28.6	0.820	0.778	0.788	0.715	0.768
DeiT-Small (early fusion)	Transformer	21.6	0.812	0.768	0.774	0.706	0.756
ResNet50	CNN	47.0	0.796	0.768	0.779	0.703	0.741
ConvNeXt-Tiny	CNN	55.6	0.822	0.767	0.767	0.680	0.757
EfficientNet-B0, OCT only	Single modality	4.0	0.809	0.764	0.771	0.683	0.753
EfficientNet-B0, SLO only	Single modality	4.0	0.806	0.758	0.762	0.673	0.723
ResNet50 (early fusion)	CNN	23.5	0.746	0.747	0.759	0.688	0.745
ResNet18 (scratch)	CNN	22.3	0.773	0.717	0.725	0.666	0.706

Table 11: Sweep 1: 13 configurations, plain recipe, 15 epochs, validation-based epoch selection. Params = total parameters of both branches. "TTA AUC" = with horizontal-flip test-time averaging.

Configuration	Val AUC	Test AUC	95% CI	TTA AUC	ES-AUC race	Worst-group
EfficientNet-B0 + recipe	0.844	0.799	[0.779, 0.820]	0.802	0.774	0.782
seed 1	0.818	0.795	[0.774, 0.815]	0.801	0.747	0.782
seed 2	0.835	0.800	[0.778, 0.821]	0.805	0.755	0.777
12 epochs	0.836	0.777	[0.757, 0.798]	0.787	0.760	0.766
+ mixup/CutMix	0.839	0.800	[0.780, 0.819]	0.805	0.765	0.786
+ drop-path 0.2, dropout 0.5	0.852	0.817	[0.797, 0.839]	0.820	0.793	0.803
lower LR (2e-4)	0.833	0.790	[0.767, 0.812]	0.803	0.779	0.783
gated fusion + modality dropout	0.832	0.810	[0.791, 0.830]	0.814	0.801	0.803
EfficientNet-B2	0.839	0.796	[0.778, 0.817]	0.806	0.765	0.783
EfficientNet-B3	0.821	0.779	[0.759, 0.799]	0.789	0.738	0.757
RegNetY-016	0.831	0.807	[0.785, 0.828]	0.812	0.773	0.803
DenseNet121	0.828	0.781	[0.760, 0.801]	0.791	0.739	0.768
Drop-path config + adversarial race debiasing	0.847	0.802	[0.781, 0.821]	0.807	0.782	0.796

Table 12: Sweep 3: EfficientNet-B0 regularisation and backbone neighbours, 25 epochs unless stated. The drop-path configuration had the best validation AUC (0.852) and was used for the cross-validation and full-data members of the final ensemble.

- **Compute.** Single runs took about 3 minutes (ResNet18) to 18 minutes (ConvNeXt-Tiny, 15 epochs) in sweep 1, and about 11 to 22 minutes in later sweeps, on a P100. EfficientNet-B0 was the cheapest strong model.